

Supplementary Legends for: Computational Tool Choice Impacts CRISPR Spacer-Protospacer Detection

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1 Supplementary Materials

This legend-only version preserves the section structure, captions, and explanatory notes from `supplementary.qmd` while omitting embedded tables and figures. Source data remain available in `draft/supp/figures/` and `draft/supp/figures/tables/`.

1.1 Tables

1.1.1 Supplementary Table S1: Tool Configuration Details

Parameters chosen for each tool to maximize recall on short spacers. BLASTn settings emphasize ungapped alignments and permissive identity/coverage to capture all valid matches; bowtie variants, minimap2, MM-seqs2, mummer4, sassy, strobealign, and x_mapper entries list versions and command-line templates. Full table is available in `draft/supp/figures/tables/supp_table_S1_tool_config.tsv`.

1.1.2 Supplementary Table S2: IMG/VR4 Contig Dataset Statistics and Filtering Steps

Summary of IMG/VR4 v1.1 high-confidence viral contigs after taxonomic filtering and length cutoffs, including the final HQ benchmark subset (`fraction_1`). See `draft/supp/figures/tables/supp_table_S2_contig_stats.tsv` for the full table.

1.1.3 Supplementary Table S3: CRISPR Spacer Dataset Composition and Statistics

Statistics for the iPHoP spacer dataset before and after complexity filtering, including length ranges, GC content, and removal criteria. Data are provided in `draft/supp/figures/tables/supp_table_S3_spacer_stats.tsv`.

1.1.4 Supplementary Table S4: Detailed Statistics for All Datasets

A. Real datasets — Counts and length/GC summaries for spacers and contigs across IMG/VR4 fractions (0.0005 through 1).

B. Simulated datasets — Equivalent summaries for all synthetic datasets used in the study. Full details are in `draft/supp/figures/tables/supp_table_S4_real_dataset_sequence_stats.tsv` and `supp_table_S4_simulated_dataset_sequence_stats.tsv`.

1.1.5 Supplementary Table S5: Simulation and job-generation configuration from `Prepare_all_jobs.ipynb`

Per-dataset group configuration for simulations and tool runs, noting search-space sizes, distance thresholds, and tool inclusion/exclusion. See `draft/supp/figures/tables/supp_table_S5_prepare_jobs.tsv` for the structured table.

1.1.6 Supplementary Table S6: Recall Values for Each Tool at Different Mismatch Thresholds

Values are counted per spacer-contig alignment using exact distance for each metric. `n_found` tallies tool-reported positive alignments at a given distance, `n_total` is the ground-truth positive count, and recall is `n_found / n_total`. Subsections report IMG/VR4 and synthetic datasets separately. Complete values reside in `draft/supp/figures/tables/supp_table_S6_recall_values.tsv`.

1.2 Supplementary Figures

1.2.1 Supplementary Figure S1: IMG/VR4 tool-vs-tool comparison matrices

Pairwise tool-comparison matrices for IMG/VR4 `fraction_1` at exact hamming distances (0-3). Each cell indicates regions detected by one tool but not another; diagonals list per-tool totals. Original figure: `draft/supp/figures/performance/imgvr4_fraction1_tool_comparison_matrix.svg`.

1.2.2 Supplementary Figure S2: Simulated dataset recall vs spacer occurrence frequency

High-occurrence-focused recall view for the high-insertion simulated dataset (`ns_500_nc_5000_HIGH_INSERTION_RATE`) at hamming distance 3. Original figure: `draft/supp/figures/performance/imgvr4_high_ins_recall_vs_occurrence.svg`.

1.2.3 Supplementary Figure S3: IMG/VR4 `fraction_1` recall vs occurrence frequency at hamming distance 3

Recall versus spacer occurrence for the real IMG/VR4 `fraction_1` dataset at hamming distance 3. Original figure: `draft/supp/figures/performance/imgvr4_fraction1_recall_vs_occurrence.png`.

1.2.4 Supplementary Figure S4: Comparison of simulated vs real spacer characteristics

Distributions comparing simulated and real spacer features (k-mer repeatability, entropy, base frequencies, GC, LCC). Panel A shows filtered iPHoP spacers; Panel B shows fully simulated spacers (`ns_100000_nc_20000`). Original figures: `figures/supp/filtered_iphop_spacers_attribute_distributions_multiplo` and `figures/supp/simulated_spacer_distributions.svg`.

1.2.5 Supplementary Figure S5: Spacer characteristic distributions by dataset

Spacer and contig characteristic distributions for real (IMG/VR4) and simulated datasets, covering spacer size, mismatch counts, and occurrence rates. Outliers were removed for `fraction_1` length visualization. Original figure: `figures/supp/real_data_contig_distributions.svg`.

1.2.6 Supplementary Figure S6: Distribution of matched spacers from `fraction_1`

Distributions of matched spacers (hamming 3) across length, number of occurrences, and per-distance alignment counts. Original figure: `figures/supp/fraction1_matched_spacers_distributions.png`.

1.2.7 Supplementary Figure S7: Peak memory scaling (real and simulated datasets)

Peak memory usage (RSS) scaling with dataset size. Panel A: IMG/VR4 subsamples; Panel B: simulated datasets. Axes are log-scaled; tool identity encoded by color/shape. Original figure: `figures/resource_usage/resource_usage_memory_2panel.svg`.

1.2.8 Supplementary Figure S8: Tool recall consistency across dataset sizes

Per-tool recall consistency at hamming distance 3. Panel A: real IMG/VR4 subsamples. Panel B: synthetic datasets. Panel C: heatmap of per-tool recall across simulated runs. Original figures: `figures/performance/imgvr4_across_fractions_recall_consistency_hamming_le3.svg`, `figures/performance/simulated_cross_fraction_consistency.svg`, and `figures/performance/simulated_heatmap_d`.

1.2.9 Supplementary Figure S9: Per-simulation non-planned match counts (Hamming vs Edit)

Grouped bars for each simulation (ns_50000_nc_5000, ns_75000_nc_5000, ns_100000_nc_10000, ns_100000_nc_20000) comparing hamming and edit distance non-planned match counts at thresholds 0–5. Original figures: [figures/ns_50000_nc_5000_background_counts_foldchange.svg](#), [figures/ns_75000_nc_5000_background_counts_foldchange.svg](#), [figures/ns_100000_nc_10000_background_counts_foldchange.svg](#), and [figures/ns_100000_nc_20000_background_counts_foldchange.svg](#).

1.3 Supplementary Notes

1.3.1 Supplementary Note 1: Coordinate Tolerance Matching Example

Illustrative case from a synthetic dataset (ns_100000_nc_10000) where BLASTn, MMseqs2, and Sassy report slightly different coordinates for the same spacer–contig locus. BLASTn aligns positions 5003–5036 with 0 mismatches; MMseqs2 reports 5002–5036 with 1 mismatch; Sassy reports 5000–5036 with a 1 bp insertion (CIGAR 34=1I1=); the planned occurrence is 5001–5036. All represent acceptable matches at maximum hamming distance 1, motivating a 5 bp coordinate tolerance when aggregating alignments.

1.3.2 Supplementary Note 2: Non-planned Match Rate Analysis

1.3.2.1 Hamming vs edit distance: per-simulation results

Per-simulation comparisons of non-planned match counts under hamming versus edit distance for four fully synthetic datasets (ns_50000_nc_5000, ns_75000_nc_5000, ns_100000_nc_10000, ns_100000_nc_20000) at thresholds 0–5. Detailed counts are in [distance_metric_analysis_v3.ipynb](#) and associated CSV exports.

1.3.2.2 Search-space-normalized non-planned match rates

1.3.2.2.1 Supplementary Table S7: Search-space-normalized metrics at hamming/edit distance 3

Normalized non-planned match rates (per spacer per Gbp and per Gbp²) for each fully synthetic simulation. Edit distance 3 produces roughly 6–8× more non-planned matches than hamming 3. Full numeric table: [draft/supp/figures/tables/search_space_normalized_metrics_threshold3.csv](#).

1.3.2.2.2 Supplementary Table S8: Semi-synthetic non-planned match rates by hamming threshold

Cumulative non-planned matches in the semi-synthetic dataset across hamming thresholds (0–5), with rates normalized by spacer and contig sequence space. Values are provided in [draft/supp/figures/tables/supp_table_S8_semi_synthetic_matches.tsv](#) and the corresponding TSV.

1.3.2.3 Caveats and interpretation

Non-planned match estimates reflect compositional differences between real and simulated sequences; semi-synthetic rates (real spacers × simulated contigs) exceed fully synthetic rates. For high thresholds (4, 5) the counts are lower bounds because only heuristic tools were feasible. These rates are most informative for relative comparisons rather than absolute prediction.

Complete analysis available in: [distance_metric_analysis_v3.ipynb](#), which includes per-distance breakdowns, tool-specific validation, and detailed comparison tables.

1.4 References and Data Availability

Analysis notebooks, source code, and generated data are available in the project repository: <https://github.com/UriNeri/space>

Key notebooks: `spacer_inspection.ipynb`, `distance_metric_analysis_v3.ipynb`, `Prepare_all_jobs.ipynb`, `Performance_simulated_v2.ipynb`, `Performance_imgvr4_v3.ipynb`.

Datasets: filtered IMG/VR4 HQ contigs (fraction_1 subset), filtered iPHoP spacers, all synthetic datasets and ground truth files (Zenodo DOI links in the main text).